

*Non-Calculator***Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.****You have 30 minutes.**_____
Signature**SHOW ALL WORK FOR FULL CREDIT!**

Find the product. Write the product in rectangular form, using exact values. (3 points)

1. $\left[\sqrt{6}(\cos 165^\circ + i \sin 165^\circ) \right] \left[\sqrt{2}(\cos 255^\circ + i \sin 255^\circ) \right]$

Convert to trigonometric form, then divide. Write the quotient in rectangular form, using exact values. (3 points)

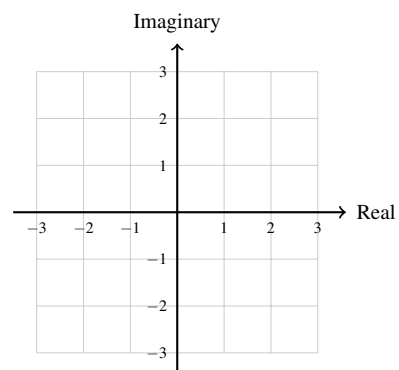
2. $\frac{-\sqrt{3} + i}{\sqrt{2} - i\sqrt{6}}$

Find the power. Write the answer in rectangular form, using exact values. (3 points)

3. $(-1 - i)^3$

Find the fourth roots of the number. Leave answers in trig form. Graph each root as a vector in the complex plane. (3 points)

4. $-2 + 2i\sqrt{3}$



The fractal called the **Mandelbrot set**: to determine if a complex number $z = a + bi$ is in this set, repeatedly compute $z, z^2 + z, (z^2 + z)^2 + z, [(z^2 + z)^2 + z]^2 + z, \dots$. The complex number z does **not** belong to the Mandelbrot set if any of the resulting absolute values exceeds 2. Otherwise z is in the set. Determine whether or not each number belongs to the Mandelbrot set. (3 points)

5. a) $z = i$	YES / NO	b) $z = 1 + i$	YES / NO

ANSWER KEY – Non-Calculator

SHOW ALL WORK FOR FULL CREDIT!

Every line shown. Point splits are an estimate of partial credit (halves are possible).

Find the product. Write the product in rectangular form, using exact values. (3 points)

Problem: $\left[\sqrt{6}(\cos 165^\circ + i \sin 165^\circ) \right] \left[\sqrt{2}(\cos 255^\circ + i \sin 255^\circ) \right]$

1. Multiply the r 's, add the θ 's.

$$r = \sqrt{6} \cdot \sqrt{2} = \sqrt{12} = 2\sqrt{3} \quad \theta = 165^\circ + 255^\circ = 420^\circ - 360^\circ = 60^\circ$$

$$2\sqrt{3}(\cos 60^\circ + i \sin 60^\circ) = 2\sqrt{3}\left(\frac{1}{2}\right) + 2\sqrt{3}i\left(\frac{\sqrt{3}}{2}\right) = \boxed{\sqrt{3} + 3i}$$

Trap: 165° and 255° aren't on the unit circle, but the sum is. Simplify $\sqrt{12} = 2\sqrt{3}$, and $\sqrt{3} \cdot \sqrt{3} = 3$.

1 pt r, 1 pt reduced angle, 1 pt exact rectangular.

Convert to trigonometric form, then divide. Write the quotient in rectangular form, using exact values. (3 points)

Problem: $\frac{-\sqrt{3} + i}{\sqrt{2} - i\sqrt{6}}$

2. Convert the top and the bottom separately (sketch each for the quadrant).

$$\text{Top: } r = \sqrt{(-\sqrt{3})^2 + 1^2} = \sqrt{4} = 2 \quad \tan \theta = \frac{1}{-\sqrt{3}} \Rightarrow \text{ref } 30^\circ, \text{ QII} \Rightarrow 150^\circ$$

$$\text{Bottom: } r = \sqrt{(\sqrt{2})^2 + (-\sqrt{6})^2} = \sqrt{8} = 2\sqrt{2} \quad \tan \theta = \frac{-\sqrt{6}}{\sqrt{2}} = -\sqrt{3} \Rightarrow \text{ref } 60^\circ, \text{ QIV} \Rightarrow 300^\circ$$

$$\begin{aligned} \frac{2 \text{ cis } 150^\circ}{2\sqrt{2} \text{ cis } 300^\circ} &= \frac{1}{\sqrt{2}} \text{ cis}(150^\circ - 300^\circ) = \frac{\sqrt{2}}{2} \text{ cis}(-150^\circ) \xrightarrow{+360^\circ} \frac{\sqrt{2}}{2} \text{ cis } 210^\circ \\ &= \frac{\sqrt{2}}{2} \left(-\frac{\sqrt{3}}{2}\right) + \frac{\sqrt{2}}{2} i \left(-\frac{1}{2}\right) = \boxed{-\frac{\sqrt{6}}{4} - \frac{\sqrt{2}}{4} i} \end{aligned}$$

Trap: rationalize $\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$. Negative angle: add 360°.

Find the power. Write the answer in rectangular form, using exact values. (3 points)

Problem: $(-1 - i)^3$

3. Convert, then De Moivre: $[r \text{ cis } \theta]^n = r^n \text{ cis}(n\theta)$.

$$r = \sqrt{(-1)^2 + (-1)^2} = \sqrt{2} \quad \tan \theta = \frac{-1}{-1} = 1, \text{ QIII} \Rightarrow 225^\circ$$

$$\begin{aligned} (\sqrt{2} \text{ cis } 225^\circ)^3 &= (\sqrt{2})^3 \text{ cis } 3(225^\circ) = 2\sqrt{2} \text{ cis } 675^\circ \xrightarrow{-360^\circ} 2\sqrt{2} \text{ cis } 315^\circ \\ &= 2\sqrt{2} \left(\frac{\sqrt{2}}{2}\right) + 2\sqrt{2} i \left(-\frac{\sqrt{2}}{2}\right) = \boxed{2 - 2i} \end{aligned}$$

Trap: $(\sqrt{2})^3 = \sqrt{2} \cdot \sqrt{2} \cdot \sqrt{2} = 2\sqrt{2}$. Both parts negative is QIII (225°), not the 45° that $\tan^{-1}$ gives.

Find the fourth roots of the number. Leave answers in trig form. Graph each root as a vector in the complex plane. (3 points)

Problem: $-2 + 2i\sqrt{3}$

4. Convert, then the n th root theorem: $\sqrt[n]{r} \operatorname{cis}\left(\frac{\theta}{n} + \frac{360^\circ k}{n}\right)$.

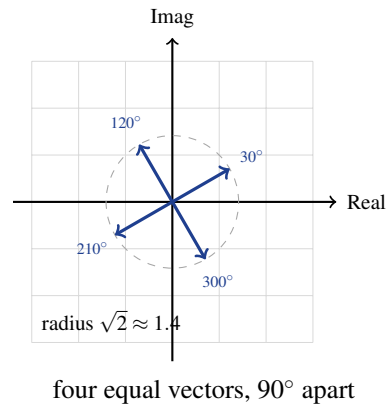
$$r = \sqrt{(-2)^2 + (2\sqrt{3})^2} = \sqrt{4 + 12} = 4$$

$$\tan \theta = \frac{2\sqrt{3}}{-2} = -\sqrt{3} \Rightarrow \text{ref } 60^\circ, \text{ QII} \Rightarrow 120^\circ$$

$$\sqrt[4]{4} = \sqrt{2} \quad \text{first } \frac{120^\circ}{4} = 30^\circ \quad + 90^\circ$$

$$\sqrt{2} \operatorname{cis} 30^\circ, \sqrt{2} \operatorname{cis} 120^\circ, \sqrt{2} \operatorname{cis} 210^\circ, \sqrt{2} \operatorname{cis} 300^\circ$$

Trap: $\sqrt[4]{4} = \sqrt{2}$ (the 4th root of 4 is the square root of 2). Stop at four roots.



The fractal called the **Mandelbrot set**: to determine if a complex number $z = a + bi$ is in this set, repeatedly compute z , $z^2 + z$, $(z^2 + z)^2 + z$, $[(z^2 + z)^2 + z]^2 + z, \dots$. The complex number z does **not** belong to the Mandelbrot set if any of the resulting absolute values exceeds 2. Otherwise z is in the set. Determine whether or not each number belongs to the Mandelbrot set. (3 points)

Problem: a) $z = i$

YES / NO

b) $z = 1 + i$

YES / NO

5. Keep computing until an absolute value goes over 2 (then NO) or the values start repeating (then YES). $|a + bi| = \sqrt{a^2 + b^2}$.

a) $z = i$: $|i| = 1$

$$z^2 + z = i^2 + i = -1 + i, \quad |-1 + i| = \sqrt{1 + 1} = \sqrt{2} \approx 1.4$$

$$(z^2 + z)^2 + z = (-1 + i)^2 + i = (1 - 2i + i^2) + i = -2i + i = -i, \quad |-i| = 1$$

$$(-i)^2 + i = -1 + i \text{ again: it repeats } (-1 + i, -i, -1 + i, \dots) \text{ and never passes } 2$$

YES

b) $z = 1 + i$: $|1 + i| = \sqrt{2} \approx 1.4 \leq 2$

$$z^2 + z = (1 + i)^2 + (1 + i) = (1 + 2i + i^2) + 1 + i = 2i + 1 + i = 1 + 3i$$

$$|1 + 3i| = \sqrt{1 + 9} = \sqrt{10} \approx 3.2 > 2 \quad \text{NO}$$

Trap: square the whole number: $(1 + i)^2 = 1 + 2i + i^2 = 2i$, not $1 + i^2$. Stop the moment one value exceeds 2.

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Signature**SHOW ALL WORK FOR FULL CREDIT!**

Find the product. Write the product in rectangular form, using exact values. (3 points)

1. $\left[\sqrt{2} \left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right) \right] \left[\sqrt{2} \left(\cos \frac{11\pi}{12} + i \sin \frac{11\pi}{12} \right) \right]$

Find the quotient. Write it in rectangular form, using exact values. (3 points)

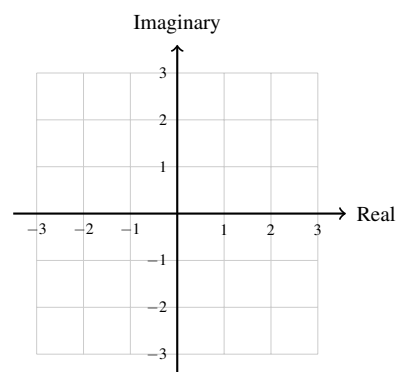
2. $\frac{\sqrt{2} + i\sqrt{6}}{-\sqrt{3} - i}$

Find the power. Write the answer in rectangular form, using exact values. (3 points)

3. $\left[\sqrt{2} \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) \right]^4$

Find the fifth roots of the number. Leave answers in trig form with θ in radians. Graph each root as a vector in the complex plane. (3 points)

4. $-1 - i\sqrt{3}$



The fractal called the **Mandelbrot set**: to determine if a complex number $z = a + bi$ is in this set, repeatedly compute z , $z^2 + z$, $(z^2 + z)^2 + z$, $[(z^2 + z)^2 + z]^2 + z, \dots$. The complex number z does **not** belong to the Mandelbrot set if any of the resulting absolute values exceeds 2. Otherwise z is in the set. Determine whether or not each number belongs to the Mandelbrot set. (3 points)

5. a) $z = -2$

YES / NO

b) $z = -1 + i$

YES / NO

ANSWER KEY – Non-Calculator

SHOW ALL WORK FOR FULL CREDIT!*Every line shown. Point splits are an estimate of partial credit (halves are possible).*

Find the product. Write the product in rectangular form, using exact values. (3 points)

Problem: $\left[\sqrt{2} \left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right) \right] \left[\sqrt{2} \left(\cos \frac{11\pi}{12} + i \sin \frac{11\pi}{12} \right) \right]$

1. Multiply the r 's, add the θ 's (common denominator first).

$$r = \sqrt{2} \cdot \sqrt{2} = 2 \quad \theta = \frac{5\pi}{4} + \frac{11\pi}{12} = \frac{15\pi}{12} + \frac{11\pi}{12} = \frac{26\pi}{12} = \frac{13\pi}{6} \xrightarrow{-2\pi} \frac{\pi}{6}$$

$$2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right) = 2 \left(\frac{\sqrt{3}}{2} + i \left(\frac{1}{2} \right) \right) = \boxed{\sqrt{3} + i}$$

Trap: given in radians, stay in radians. $2\pi = \frac{12\pi}{6}$.

Find the quotient. Write it in rectangular form, using exact values. (3 points)

Problem: $\frac{\sqrt{2} + i\sqrt{6}}{-\sqrt{3} - i}$

2. Convert top and bottom, divide the r 's, subtract the θ 's.

$$\text{Top: } r = \sqrt{(\sqrt{2})^2 + (\sqrt{6})^2} = \sqrt{8} = 2\sqrt{2} \quad \tan \theta = \frac{\sqrt{6}}{\sqrt{2}} = \sqrt{3} \Rightarrow \text{QI, } 60^\circ$$

$$\text{Bottom: } r = \sqrt{(-\sqrt{3})^2 + (-1)^2} = 2 \quad \tan \theta = \frac{-1}{-\sqrt{3}} = \frac{1}{\sqrt{3}} \Rightarrow \text{ref } 30^\circ, \text{ QIII} \Rightarrow 210^\circ$$

$$\frac{2\sqrt{2} \text{cis } 60^\circ}{2 \text{cis } 210^\circ} = \sqrt{2} \text{cis}(-150^\circ) \xrightarrow{+360^\circ} \sqrt{2} \text{cis } 210^\circ = \sqrt{2} \left(-\frac{\sqrt{3}}{2} + i \left(-\frac{1}{2} \right) \right) = \boxed{-\frac{\sqrt{6}}{2} - \frac{\sqrt{2}}{2}i}$$

Trap: both parts of the bottom negative means QIII: 210° , not 30° .

Find the power. Write the answer in rectangular form, using exact values. (3 points)

Problem: $\left[\sqrt{2} \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) \right]^4$

3. De Moivre: raise r to the 4th, multiply the angle by 4.

$$(\sqrt{2})^4 = 4 \quad 4 \cdot \frac{2\pi}{3} = \frac{8\pi}{3} \xrightarrow{-2\pi} \frac{8\pi}{3} - \frac{6\pi}{3} = \frac{2\pi}{3}$$

$$4 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) = 4 \left(-\frac{1}{2} + i \left(\frac{\sqrt{3}}{2} \right) \right) = \boxed{-2 + 2\sqrt{3}i}$$

Trap: $(\sqrt{2})^4 = 4$, not $4\sqrt{2}$. $\frac{2\pi}{3}$ is QII: cos negative, sin positive.

Find the fifth roots of the number. Leave answers in trig form with θ in radians. Graph each root as a vector in the complex plane. (3 points)

Problem: $-1 - i\sqrt{3}$

4. Convert, then roots. Radians throughout.

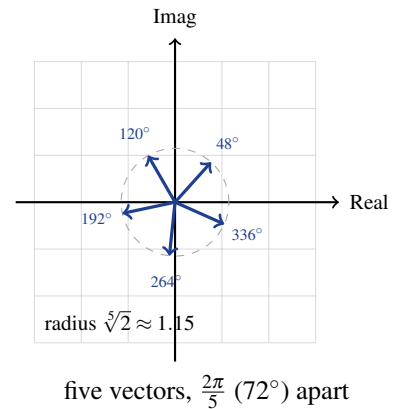
$$r = \sqrt{(-1)^2 + (-\sqrt{3})^2} = \sqrt{4} = 2$$

$$\tan \theta = \frac{-\sqrt{3}}{-1} = \sqrt{3} \Rightarrow \text{ref } \frac{\pi}{3}, \text{ QIII} \Rightarrow \frac{4\pi}{3}$$

$$\sqrt[5]{2} \quad \text{first: } \frac{4\pi/3}{5} = \frac{4\pi}{15} \quad + \frac{2\pi}{5} = + \frac{6\pi}{15}$$

$$\sqrt[5]{2} \text{ cis } \frac{4\pi}{15}, \sqrt[5]{2} \text{ cis } \frac{2\pi}{3}, \sqrt[5]{2} \text{ cis } \frac{16\pi}{15}, \sqrt[5]{2} \text{ cis } \frac{22\pi}{15}, \sqrt[5]{2} \text{ cis } \frac{28\pi}{15}$$

Trap: $\frac{10\pi}{15} = \frac{2\pi}{3}$ (simplify). $\sqrt[5]{2}$ stays a radical.



The fractal called the **Mandelbrot set**: to determine if a complex number $z = a + bi$ is in this set, repeatedly compute $z, z^2 + z, (z^2 + z)^2 + z, [(z^2 + z)^2 + z]^2 + z, \dots$. The complex number z does **not** belong to the Mandelbrot set if any of the resulting absolute values exceeds 2. Otherwise z is in the set. Determine whether or not each number belongs to the Mandelbrot set. (3 points)

Problem: a) $z = -2$

YES / NO

b) $z = -1 + i$

YES / NO

5. Keep computing until an absolute value goes over 2 (then NO) or the values start repeating (then YES).

a) $z = -2$: $|-2| = 2$ (equal to 2 does not "exceed" 2, keep going)

$$z^2 + z = 4 - 2 = 2, \quad |2| = 2$$

$$(2)^2 + (-2) = 2 \quad \text{again: stuck at 2 forever, never over 2} \quad \boxed{\text{YES}}$$

b) $z = -1 + i$: $|z| = \sqrt{2} \approx 1.4$

$$z^2 + z = (1 - 2i + i^2) + (-1 + i) = -2i - 1 + i = -1 - i, \quad |-1 - i| = \sqrt{2}$$

$$(z^2 + z)^2 + z = (-1 - i)^2 + (-1 + i) = (1 + 2i + i^2) - 1 + i = 2i - 1 + i = -1 + 3i$$

$$|-1 + 3i| = \sqrt{1 + 9} = \sqrt{10} \approx 3.2 > 2 \quad \boxed{\text{NO}}$$

Trap: b) passes the first two checks and fails on the third. Don't stop early and call it YES.

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Signature**SHOW ALL WORK FOR FULL CREDIT!**

Find the product. Write the product in rectangular form, using exact values. (3 points)

1. $\left[2\sqrt{3} \left(\cos \frac{13\pi}{12} + i \sin \frac{13\pi}{12} \right) \right] \left[\sqrt{3} \left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right) \right]$

Convert to trigonometric form, then divide. Write the quotient in rectangular form, using exact values. (3 points)

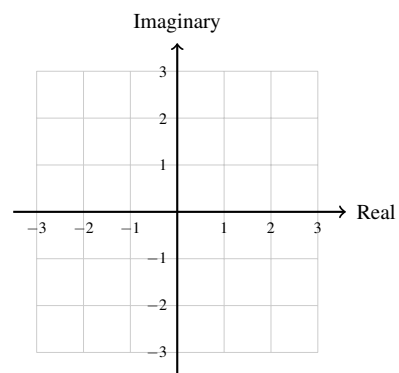
2. $\frac{-2 - 2i\sqrt{3}}{\sqrt{2} - i\sqrt{6}}$

Find the power. Write the answer in rectangular form, using exact values. (3 points)

3. $\left[\sqrt{2} \left(\cos \frac{7\pi}{12} + i \sin \frac{7\pi}{12} \right) \right]^4$

Find the cube roots of the number. Leave answers in trig form. Graph each cube root as a vector in the complex plane. (3 points)

4. $2 - 2i\sqrt{3}$



The fractal called the **Mandelbrot set**: to determine if a complex number $z = a + bi$ is in this set, repeatedly compute $z, z^2 + z, (z^2 + z)^2 + z, [(z^2 + z)^2 + z]^2 + z, \dots$. The complex number z does **not** belong to the Mandelbrot set if any of the resulting absolute values exceeds 2. Otherwise z is in the set. Determine whether or not each number belongs to the Mandelbrot set. (3 points)

5. a) $z = -i$	YES / NO	b) $z = 2i$	YES / NO

ANSWER KEY – Non-Calculator

SHOW ALL WORK FOR FULL CREDIT!*Every line shown. Point splits are an estimate of partial credit (halves are possible).*

Find the product. Write the product in rectangular form, using exact values. (3 points)

Problem: $\left[2\sqrt{3} \left(\cos \frac{13\pi}{12} + i \sin \frac{13\pi}{12} \right) \right] \left[\sqrt{3} \left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right) \right]$

1. Multiply the r 's, add the θ 's (common denominator first).

$$r = (2\sqrt{3})(\sqrt{3}) = 2 \cdot 3 = 6 \quad \theta = \frac{13\pi}{12} + \frac{15\pi}{12} = \frac{28\pi}{12} = \frac{7\pi}{3} \xrightarrow{-2\pi} \frac{\pi}{3}$$

$$6 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) = 6 \left(\frac{1}{2} \right) + 6i \left(\frac{\sqrt{3}}{2} \right) = \boxed{3 + 3\sqrt{3}i}$$

Trap: $\frac{13\pi}{12}$ is not on the unit circle, but the sum is. Add first, then look.

Convert to trigonometric form, then divide. Write the quotient in rectangular form, using exact values. (3 points)

Problem: $\frac{-2 - 2i\sqrt{3}}{\sqrt{2} - i\sqrt{6}}$

2. Convert top and bottom, divide the r 's, subtract the θ 's.

$$\text{Top: } r = \sqrt{(-2)^2 + (-2\sqrt{3})^2} = \sqrt{4 + 12} = 4 \quad \tan \theta = \frac{-2\sqrt{3}}{-2} = \sqrt{3} \Rightarrow \text{ref } 60^\circ, \text{ QIII} \Rightarrow 240^\circ$$

$$\text{Bottom: } r = \sqrt{(\sqrt{2})^2 + (-\sqrt{6})^2} = \sqrt{8} = 2\sqrt{2} \quad \tan \theta = \frac{-\sqrt{6}}{\sqrt{2}} = -\sqrt{3} \Rightarrow \text{ref } 60^\circ, \text{ QIV} \Rightarrow 300^\circ$$

$$\frac{4 \text{cis } 240^\circ}{2\sqrt{2} \text{cis } 300^\circ} = \frac{2}{\sqrt{2}} \text{cis}(-60^\circ) \xrightarrow{+360^\circ} \sqrt{2} \text{cis } 300^\circ = \sqrt{2} \left(\frac{1}{2} \right) + \sqrt{2}i \left(-\frac{\sqrt{3}}{2} \right) = \boxed{\frac{\sqrt{2}}{2} - \frac{\sqrt{6}}{2}i}$$

Trap: $\frac{2}{\sqrt{2}} = \sqrt{2}$. Both parts of the top negative: QIII, 240° (not 60°).

Find the power. Write the answer in rectangular form, using exact values. (3 points)

Problem: $\left[\sqrt{2} \left(\cos \frac{7\pi}{12} + i \sin \frac{7\pi}{12} \right) \right]^4$

3. De Moivre: raise r to the 4th, multiply the angle by 4, then reduce.

$$(\sqrt{2})^4 = 4 \quad 4 \cdot \frac{7\pi}{12} = \frac{28\pi}{12} = \frac{7\pi}{3} \xrightarrow{-2\pi} \frac{\pi}{3}$$

$$4 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) = 4 \left(\frac{1}{2} \right) + 4i \left(\frac{\sqrt{3}}{2} \right) = \boxed{2 + 2\sqrt{3}i}$$

Trap: $\frac{7\pi}{12}$ is not a unit-circle angle, but $4 \cdot \frac{7\pi}{12}$ reduces to one. Multiply first, then reduce.

Find the cube roots of the number. Leave answers in trig form. Graph each cube root as a vector in the complex plane. (3 points)

Problem: $2 - 2i\sqrt{3}$

4. Convert, then cube roots. $\sqrt[3]{4}$ does not simplify, so leave it.

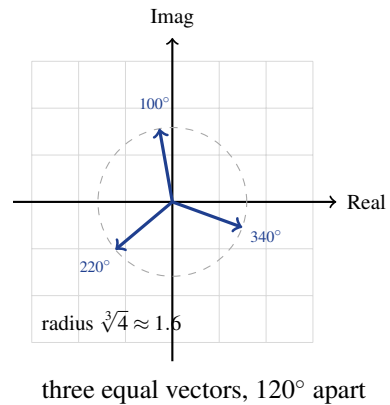
$$r = \sqrt{2^2 + (-2\sqrt{3})^2} = \sqrt{4 + 12} = 4$$

$$\tan \theta = \frac{-2\sqrt{3}}{2} = -\sqrt{3} \Rightarrow \text{ref } 60^\circ, \text{ QIV} \Rightarrow 300^\circ$$

$$\sqrt[3]{4} \quad \text{first: } \frac{300^\circ}{3} = 100^\circ \quad + 120^\circ$$

$$\boxed{\sqrt[3]{4} \text{cis } 100^\circ, \sqrt[3]{4} \text{cis } 220^\circ, \sqrt[3]{4} \text{cis } 340^\circ}$$

Trap: divide the angle, not the radius: $\sqrt[3]{4}$, not $\frac{4}{3}$.



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Problem: a) $z = -i$

YES / NO

b) $z = 2i$

YES / NO

5. Keep computing until an absolute value goes over 2 (then NO) or the values start repeating (then YES).

a) $z = -i$: $|-i| = 1$

$$z^2 + z = (-i)^2 - i = -1 - i, \quad |-1 - i| = \sqrt{2} \approx 1.4$$

$$(z^2 + z)^2 + z = (-1 - i)^2 - i = (1 + 2i + i^2) - i = 2i - i = i, \quad |i| = 1$$

$$i^2 - i = -1 - i \text{ again: it repeats } (-1 - i, i, -1 - i, \dots) \text{ and never passes 2}$$

YES

b) $z = 2i$: $|2i| = 2$ (not over 2 yet, keep going)

$$z^2 + z = (2i)^2 + 2i = -4 + 2i, \quad |-4 + 2i| = \sqrt{16 + 4} = \sqrt{20} \approx 4.5 > 2$$

NO

Trap: $(-i)^2 = i^2 = -1$ and $(2i)^2 = 4i^2 = -4$. Exactly 2 is still in; only more than 2 is out.